

Rishi Ravula

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EDUCATION

Duke University

Bachelor of Science Engineering, Electrical Computer Engineering & Computer Science

August 2020 - May 2024

GPA: 3.78

CERTIFICATIONS & AWARDS

AWS Solutions Architect Associate, Capital One Enterprise-Wide Hackathon Winner (1/700). HackDuke 2021 Health Sector Winner (1/300)

PROFESSIONAL EXPERIENCE

Coinbase

Software Engineer

New York, NY, USA

October 2025 - Present

- Built 40+ per-service LLM agents, **reducing production triage time by 75%** and implementing over 100+ Payment Platform feature requests
- Developed an AI code-intelligence bank indexing 2,700+ projects, accelerating code discovery, and agentic development for 3,000+ engineers.
- Founded Paygent, a payments AI orchestration harness reducing agent bootstrapping from 2 weeks to 2 days across 15+ internal AI use cases.
- Created CBGuard, an AI circuit-breaker layer that blocks unsafe tool calls, caps runaway agents, and **reduced agent execution fails by 72%**.

Capital One

Senior Associate Software Engineer

New York, NY, USA

August 2024 - September 2025

- Created a **MCP-based LLM Agents** to perform continuous, near-real-time monitoring: **decreasing Mean Time to Acknowledge to minutes**.
- Built low-latency filtering APIs over DynamoDB, enabling real-time queries on 400+ datasets; **\$1M+ in data monetization at 60K+ QPS**.
- Spearheading MVP for vector and graph databases, enabling low-latency access in read-distributed systems; developing from 0 to 1.
- Integrated **Glue ETL pipelines and LLMs** in recruitment systems, achieving **90% accuracy** and **reducing human bias by 37%**.
- Built an automated data loader using S3, Glue, and Lambda, to **restore and recover TBs of data during disasters; cut recovery time by 90%**.

Natter

Software Engineer Contract

London, UK

August 2023 - January 2024

- Built a highly available, scalable video conferencing platform supporting **10,000+ concurrent users** under heavy traffic.
- Developed **fine-tuned, open-source LLMs**, trained on proprietary data, utilizing zero-shot inferencing and novel embedding techniques.
- Increased attentive engagement by 40%** through AI-driven user matching and predictive analytics, enhancing interaction quality.

Capital One

Software Engineering Intern

New York, USA

May 2023 - August 2023

- Conducted the migration of a core account, Java Spring Boot service from an EC2 instance to a serverless Fargate infrastructure.
- Migration **outperforms the legacy service 45% faster in response times** and tolerates **2.6 million monthly transactions** in production.
- Orchestrated Fargate clusters, Load Balancers, IAM roles and more, including cross-region failovers and Docker containerization.

Tuned.com

Software Engineering Intern

Scottsdale, AZ, USA

May 2022 - August 2022

- Developed a high-frequency trading alert system for the crypto automated trading platform using JavaScript, Kotlin, and GraphQL.
- Alerts were sent as client-facing notifications with low-latency routing mechanisms and with up to **an 85% success rate per execution**.
- Contributed to the platform's success prior to its acquisition, alerts supplementing over **1 million trades** and **\$6 billion in transaction volume**.

PROJECTS

Promptful.ai

- Building Promptful, a full-stack app to concurrently compare LLM outputs from OpenAI, Claude, etc. using Go and Typescript.
- Leading 3 developers to eliminate the friction of prompt testing across LLMs by centralizing access, state management, model comparison.
- Integrated Hugging Face Transformers and TRL to support user low-code fine-tuning and local inference with Dockerized self-hosted models.
- Built Go-based backend microservice and Typescript UI for model comparison, prompt tuning, and personalized usage feedback.

Capital One Enterprise-Wide Hackathon

- Architected and led a team to develop a RAG-powered Slack/VSCode agent using Docker, pgvector, and LLaVa models.
- Unanimously won 1st place; solution adopted for production by Director of Slack with me as team lead overseeing implementation.
- Estimated to save \$1M during production, increasing question answer search accuracy by 80% and impacting 30,000 employees.

AI Celebrity Replica

- Fine-tuned Phi-3 and Mistral 8B for celebrity emulation: LangFuse, PostgreSQL and Tavily for 40% response accuracy improvement.
- Orchestrated AWS services (EC2, ELB, Route 53, Docker) for 99.9% uptime/scalability to enable retrieval augmented generation.
- Sold for 6 figures and delivered to Sensay, securing 2 full-time roles and a 30% engagement boost.

SKILLS

Languages: Python, Go, TypeScript, Scala, Kotlin, Java, SQL, Swift, C, HTML, CSS

Technologies & Tools: AWS, Docker, Terraform, Kubernetes, LLMs, GraphQL, REST APIs, Redis, PostgreSQL, Jenkins, Git,

Libraries & Frameworks: PyTorch, FastAPI, Numpy, Pandas, LangGraph, LangChain, HuggingFace Transformers, TRL